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Design of polarization-independent 1 × 2 optical power splitter based on silicon nitride slot waveguide structure

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This paper introduces a novel design of a three-layer slot waveguide structure, serving as a polarization-independent optical power splitter based on Si/SiN_x/Si materials. This design addresses the common challenges of structural complexity and high loss typically encountered in most polarization-independent optical power splitters. It achieves successful 1 × 2 power splitting for 1550 nm optical signals. The refractive index of the SiN_x layer is tunable through the ion-assisted deposition method, which enables the alignment of output image points for both transverse electric (TE) and transverse magnetic (TM) modes, thereby achieving polarization-independent performance. The width of the Multimode Interference (MMI) region is reasonably selected to balance between device size and loss ensuring optimal performance. Additionally, an optimized tapered waveguide design has been incorporated to effectively reduce insertion loss. The device is simulated using the Frequency Domain Finite Difference (FDE) method and the Eigenmode Expansion (EME) method. Simulation results show that the coupling region of the device measures 4 μm×15.6 μm, with insertion losses for both TE and TM polarization modes below 0.034 dB at 1550 nm, and below 0.175 dB over a 100 nm bandwidth. This optical power splitter design holds significant potential for future integrated optical path applications.

The rapid advancement of optical communication technologies has led to an increasing demand for high-speed, high-bandwidth data transmission, necessitating photonic integrated circuits (PICs)[1] that offer exceptional performance and reliability. Within PICs, optical power splitters enable the realization of various complex photonic devices, including multiplexers[2], optical switches[3], and optical modulators[4], among others. Currently, the primary device structures for optical power splitters include Y-branch type[5,6], adiabatic coupler[7], directional coupler (DC)[8,9] and multimode interference (MMI)[10,11] types. Among these components, the multimode interference coupler (MMI) has emerged as a key photonic device due to its excellent coupling efficiency, low insertion loss, and significant design flexibility. In recent years, silicon nitride (SiN) materials[12–15] have become the material of choice for high-performance photonic devices, owing to their superior optical properties, broad spectral transmittance, and cost-effectiveness.

However, traditional multimode couplers often suffer from polarization dependence, leading to reduced signal transmission efficiency across varying polarization states. Consequently, current research is focused on developing couplers that can achieve polarization independence. Studies have shown that reasonable design and optimization can minimize the polarization-induced transmission loss, significantly enhancing overall coupler performance. Polarization-independent functionality can typically be achieved through the application of a metal overlay[16], adjustment of the MMI waveguide width[17], or the use of subwavelength grating (SWG) structures[18,19]. However, metal coatings are additional device losses, adjusting the MMI waveguide width narrows the bandwidth and reduces tolerance to manufacturing errors, limiting the device applicability. Furthermore, SWG structures can be complex to implement. Alternatively, the slot waveguide structure[20–22] offers a practical solution for achieving polarization independence by precisely adjusting the geometric parameters and refractive index of the upper and lower material layers, while maintaining beam mode stability. This approach enhances the performance and reliability of the coupler. Recent studies have demonstrated that multi-layer structures composed of silicon nitride and other materials, such as silicon or silica, can achieve excellent coupling characteristics across a range of operating wavelengths[23].

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This study presents the design of a low-loss, polarization-independent multimode interference optical power splitter that utilizes MMI theory for power distribution. The device features a three-layer slot waveguide structure composed of Si/ SiN_x /Si, achieving polarization independence by adjusting the refractive index of the SiN_x layer. By appropriately selecting the coupling zone dimensions and introducing a conical structure in the input and output waveguides, the coupling efficiency between single-mode and multimode waveguides is enhanced, resulting in reduced device loss and improved performance. In this paper, the FDE[24] and EME[25] methods are utilized for device simulation. Furthermore, fabrication tolerance analysis demonstrates that this device exhibits excellent tolerance. Compared to similar devices, our design offers lower loss and wider bandwidth, coverage across the entire C-band, as well as parts of the S and L bands.

Theory and structural design

Theory

After propagating a certain distance, the incident light signal can periodically exhibit a light intensity distribution within the waveguide that resembles the incident light field, a phenomenon known as the self-image of the input signal. This effect demonstrates the self-imaging principle of optical signals in multimode waveguides[10]. It results from the coherent interference and coupling between different propagation modes within the waveguide. In the MMI waveguide with symmetric interference, the length, L_{MMI} corresponding to the N-fold images point is given by:

$$L_{MMI} = \frac{3L_\pi}{4N} \tag{1}$$

In this study, the device is a 1 × 2 optical power splitter, meaning that the 1 × 2 optical splitting function is achieved when $N = 2$, when the length of the MMI waveguide equals L_{MMI}. The formula (1) can be expressed as:

$$L_{MMI} = \frac{3L_\pi}{8} \tag{2}$$

where L_π is the beat length of the two lowest-order modes, which represents the length of the first image point of the optical signal in the MMI waveguide. It can be expressed as:

$$L_\pi = \frac{\lambda}{\beta_0 - \beta_1} \approx \frac{4n_e W_e^2}{3\lambda_0} \tag{3}$$

where λ is the wavelength, β_0 and β_1 represent the propagation constants of the fundamental mode and the first-order mode, respectively. λ_0 is the wavelength of the incident optical signal, W_e is the equivalent width of the fundamental mode, and n_e is the effective refractive index of the waveguide.

Device structure design

Figure 1a shows the polarization-independent MMI structure designed in this study. The device consists of three main components: the input waveguide, the MMI waveguide, and the output waveguide. After traversing the multimode interference coupling region, the 1550 nm optical signal input at port 1 is split into two equal beams, which are then output from ports 2 and 3, respectively. The input/output waveguides consist of both straight and tapered waveguides. The straight waveguide has a length of 5 μm, while the tapered waveguide has a length of L_{taper}, with its width gradually varying from w_1 to W_{taper}. The input waveguide is positioned at the center of the MMI waveguide, with a specified distance between the two output waveguides is $W_{MMI}/4$. The waveguide

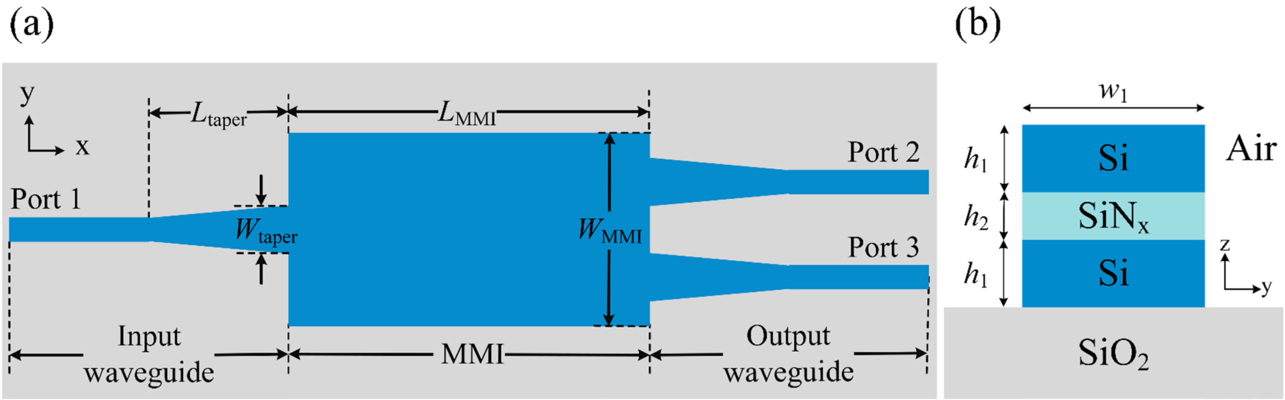

Fig. 1. Schematic diagram of the MMI optical power splitter structure (**a**) Plan of the device, the blue part shows the structure and shape of the device, the optical signal input from ports 1, output from ports 2 and 3 (**b**) The cross-section of the input/output waveguide, the blue part represents the material silicon, the light blue part is silicon nitride, and the gray part is the substrate silicon dioxide.

cross-sectional structure is shown in Fig. 1b, featuring a three-layer trough waveguide design comprising upper, middle, and lower layers. The thickness of the Si layer and SiN_x sandwich are $h_1 = 0.25$ µm and $h_2 = 0.1$ µm, respectively. At the operating wavelength of 1550 nm, the refractive index of Si is 3.47, while the substrate, SiO2, has a refractive index of 1.444. The upper cladding is air, and the refractive index of the SiN_x sandwich material, which can be controlled by Plasma-Enhanced Chemical Vapor Deposition (PECVD)[26], range from 1.72 to 3.43.

The width w_1 of the input and output straight waveguides is designed using the effective refractive index method to ensure single-mode operation, thereby minimizing transmission loss and improving the stability of the optical signal. The waveguide width is simulated using the FDE solver in lumerical MODE solutions software. Figure 2 shows how the effective refractive index (neff) increases with the width of the input straight waveguide (w_1). The first-order TE_1 mode is observed when w_1 reaches 0.85 µm, indicating that single-mode conditions are met when the width of the input and output straight waveguides w_1 is less than 0.85 µm. The cross-section design of multimode region is identical to that of the input/output waveguides, and the MMI region adopts a conventional rectangular shape. The waveguide width is W_{MMI}, and the length is L_{MMI}.

Parameter optimization

Realization of polarization-independent function

To achieve polarization-independent functionality, the device described in this study utilizes a slot waveguide structure, which results in an electric field discontinuity between materials with different refractive indices. As depicted in Fig. 3, the transverse electrical (TE) modes and transverse magnetic (TM) modes are confined to different material layers due to the refractive index contrast between Si and SiN_x materials. Specifically, TE modes

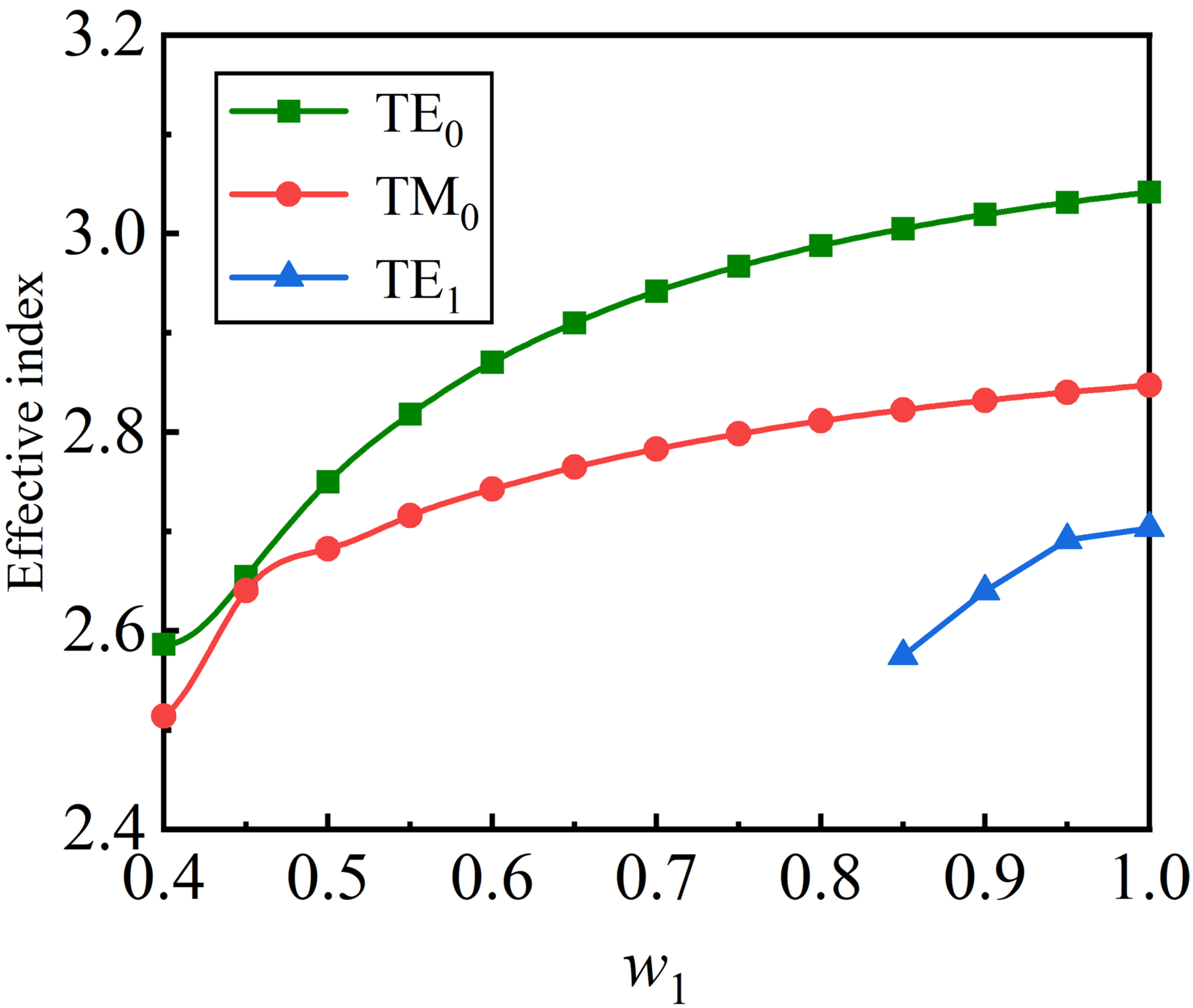

Fig. 2. Relation between the effective index and the waveguide width. The green curve is the fundamental module of TE, the red curve is the fundamental module of TM, and the blue curve is the first-order module of TE. The effective refractive index of the three modes increases gradually with the increase of the waveguide width. When w_1 is greater than or equal to 0.85 µm, TE appears in first-order mode.

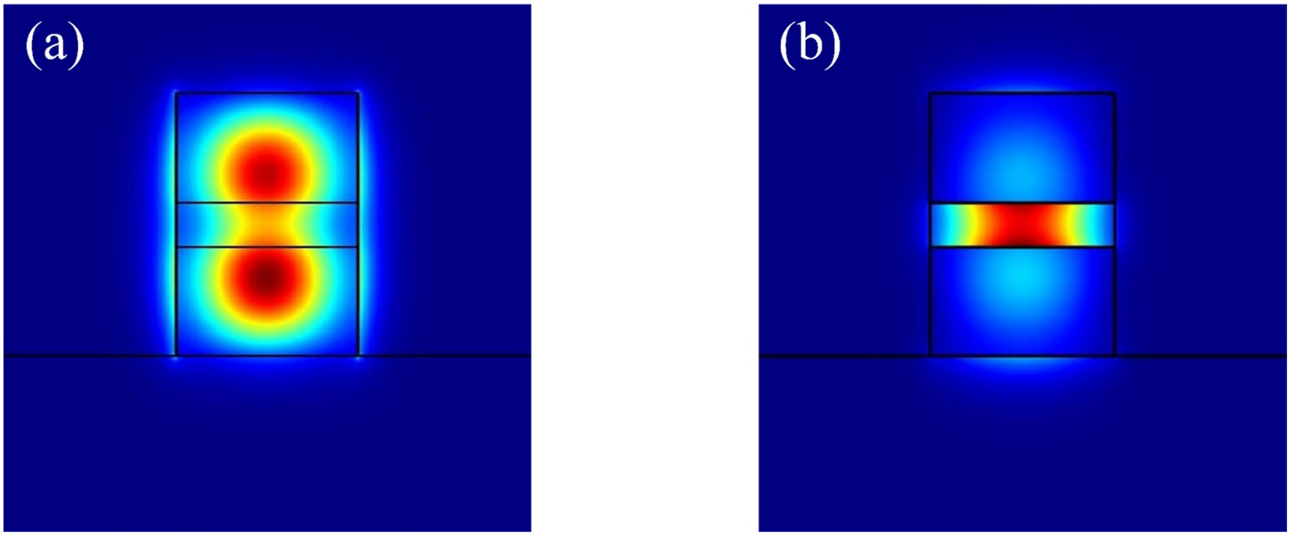

Fig. 3. Mode field distribution of light in a slot waveguide structure (**a**) TE mode is mainly confined to transmit in the upper and lower layers (**b**) TM mode is mainly confined to transmit in the middle slit.

propagate in Si, while TM modes propagate in SiN_x. To ensure polarization independence, the refractive index $n(SiN_x)$ of the SiN_x layer can be adjusted to match the beat length of the TE and TM modes. This adjustment ensures that the output image points for all polarization states of the optical signal with a wavelength of 1550 nm remain at consistent positions.

The MODE solution in lumerical software is used to simulate the 1×2 optical power splitter, and the propagation constants β_0 and β_1 for the TE mode, TM mode, and first-order mode are calculated. The β_0 and β_1 are brought into Eq. (3) to obtain the beat length for the TE and TM modes, respectively. To achieve polarization independence, the following condition must be satisfied:

$$L_{\pi(TE)} = L_{\pi(TM)} \tag{4}$$

where $L_{\pi\,(TE)}$ and $L_{\pi\,(TM)}$ are the beat lengths of the TE and TM polarization modes at wavelength of 1550 nm, respectively.

Figure 4 shows the variation of the MMI length L_{MMI} for both TE and TM modes as a function of the refractive index $n(SiN_x)$ of SiN_x, with W_{MMI} set to 3.5 μm, 4 μm, and 4.5 μm, respectively. The figure reveals that L_{MMI} increases as the refractive index of SiN_x rises, while keeping the wavelength and W_{MMI} constant. Since the TM mode propagates primarily in the SiN_x layer, whereas the TE mode is mostly confined to the Si layer, the effect of changing the SiN_x refractive index is more significant on the TM mode than on the TE mode. The beat length for the TE polarization mode increases less compared to that of the TM polarization mode. An intersection between the two curves exists, where the intersection point meets $L_{MMI(TE)} = L_{MMI(TM)}$. The refractive index of $n(SiN_x)$ at the intersection point is the value required for achieving polarization-independent operation.

MMI area optimization

The optimal MMI size is determined by optimizing the multi-mode coupling region. As shown in Fig. 4, different MMI widths correspond to different refractive indices of $n(SiN_x)$. Figure 5a presents the relationship between the length of the coupling region L_{MMI} for the two polarization states and the width W_{MMI} of the coupling region under the condition of polarization independence. The coupling length L_{MMI} for both TE and TM modes increases uniformly as the width of the coupling region W_{MMI} increases. When polarization independence is achieved, $L_{MMI(TE)} = L_{MMI(TM)}$, the curves for the TE and TM modes coincide, resulting in a single curve in the figure. Based on this, the simulations for MMIs of various sizes are conducted, and the optimal MMI width W_{MMI} is selected by considering factors such as compactness and lower loss.

Insertion loss is a critical parameter in the design of optical waveguide devices. The calculation formula is given as follows:

$$IL = -10log\left(\frac{P_2 + P_3}{P_1}\right) \tag{5}$$

where P_2 and P_3 represent the output powers of the two output ports (Port 2 and Port 3), respectively, and P_1 is the input power at Port 1.

The variation of insertion loss IL with respect to the coupling region width W_{MMI} is shown in Fig. 5b. As the device size increases, the insertion loss (IL) for both TE and TM modes decreases. Additionally, as the MMI region width (W_{MMI}) increases, the curve gradually flattens. In this study, a coupling region width of W_{MMI} =

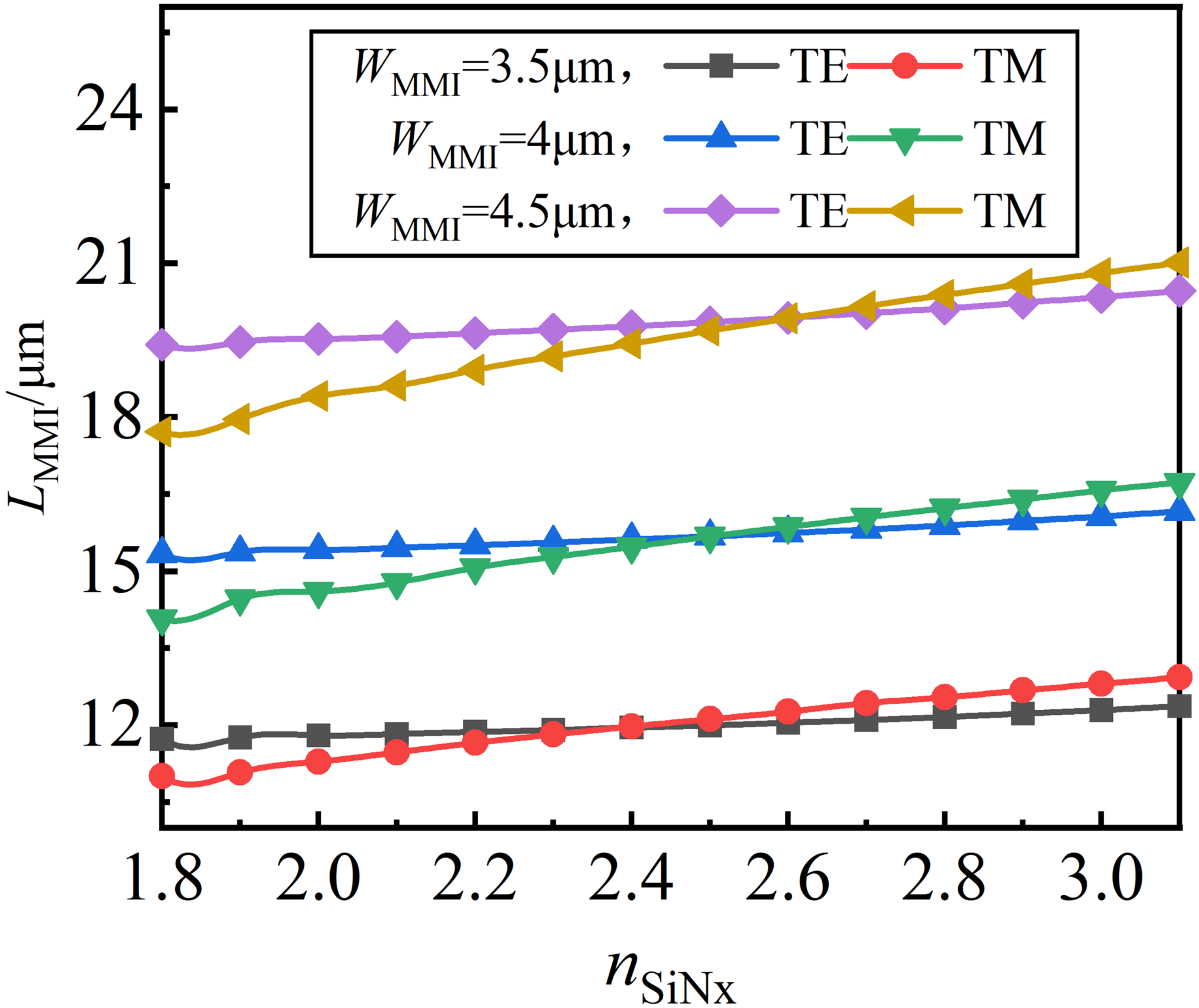

Fig. 4. The figure shows how L_{MMI} changes with $n(SiN_x)$ when W_{MMI} is 3.5, 4, and 4.5 μm, respectively. The beat length of TE mode is longer than that of TM mode when $n(SiN_x)$ is small. The beat length of both TE and TM modes grow gradually as $n(SiN_x)$ increases, with TM mode having a larger growth tendency than TE mode. The two curves intersect as $n(SiN_x)$ grows.

4 μm is selected for the device. To satisfy the single-mode condition, the width w_1 of the straight waveguide is set to 0.5 μm. The length of the coupling area L_{MMI} is 15.6 μm, and the refractive index of the sandwich layer $n(SiN_x)$ is 2.51.

Optimization of tapered waveguides

The introduction of conical structures in the input and output waveguides can significantly enhance image clarity, reduce additional device losses, improve the coupling efficiency between the single-mode input and the multi-mode interference region, and further increase the fabrication tolerance of the device. Scanning optimization is conducted on the taper length L_{taper} and width W_{taper}. As shown in Fig. 6a, the insertion loss initially increases and then decreases as the taper width W_{taper} increases. This trend occurs because, when the taper width is equal to W_{taper} =0.5 μm, the input and output waveguides are straight, resulting in higher insertion loss *IL*. With the As W_{taper} increases, *IL* gradually decreases, reaching its minimum when W_{taper} =1 μm, indicating that the taper structure effectively reduces the device's insertion loss. However, when W_{taper} > 1 μm, *IL* increases with the further increases in W_{taper}. This is because as W_{taper} increases, the distance between the two output waveguides becomes smaller, leading to coupling between the two output signals (i.e., crosstalk), which results in increased loss. Therefore, selecting an appropriate taper width is crucial for minimizing device loss. Next, the taper length L_{taper} is scanned. As shown in Fig. 6b, *IL* decreases with increasing taper length. When L_{taper} =5.5 μm, the losses for both TE and TM modes are minimized. At this point, the insertion loss *IL* of the TE polarization mode is less than 0.034 dB, while the insertion loss *IL* of the TM polarization mode is less than 0.031 dB.

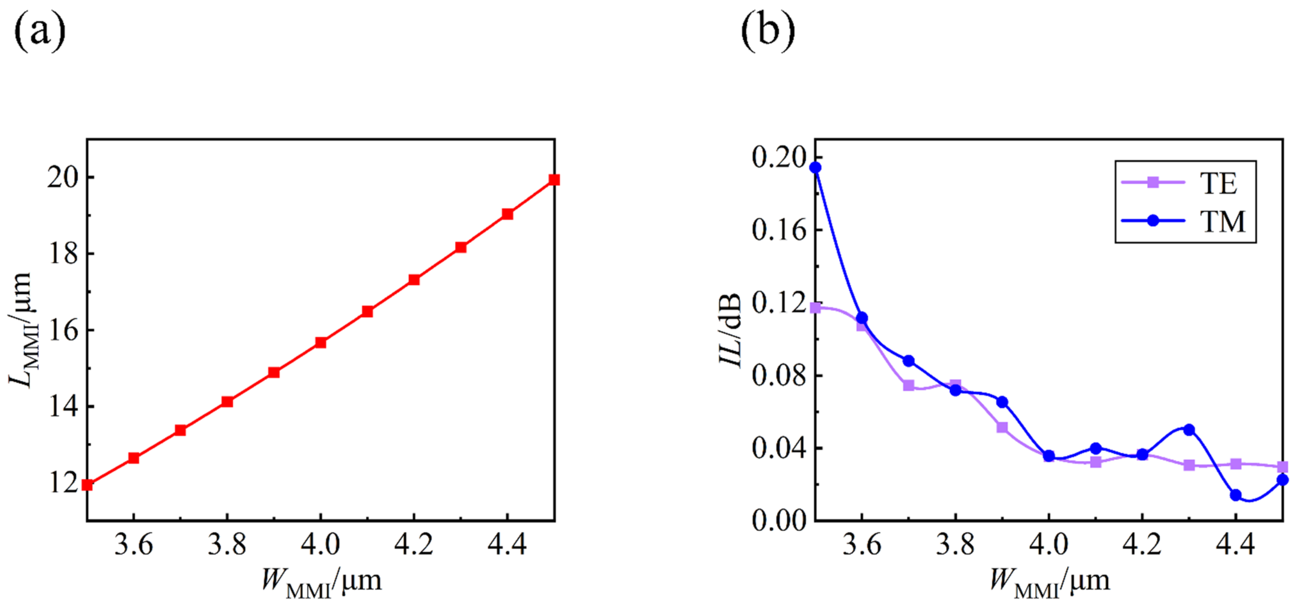

Fig. 5. Under polarization independent conditions (**a**) the change of MMI length L_{MMI} with width W_{MMI} (**b**) the change of insertion loss *IL* with W_{MMI}. With the increase of W_{MMI}, the insertion loss gradually decreases, and the trend is gradually flat.

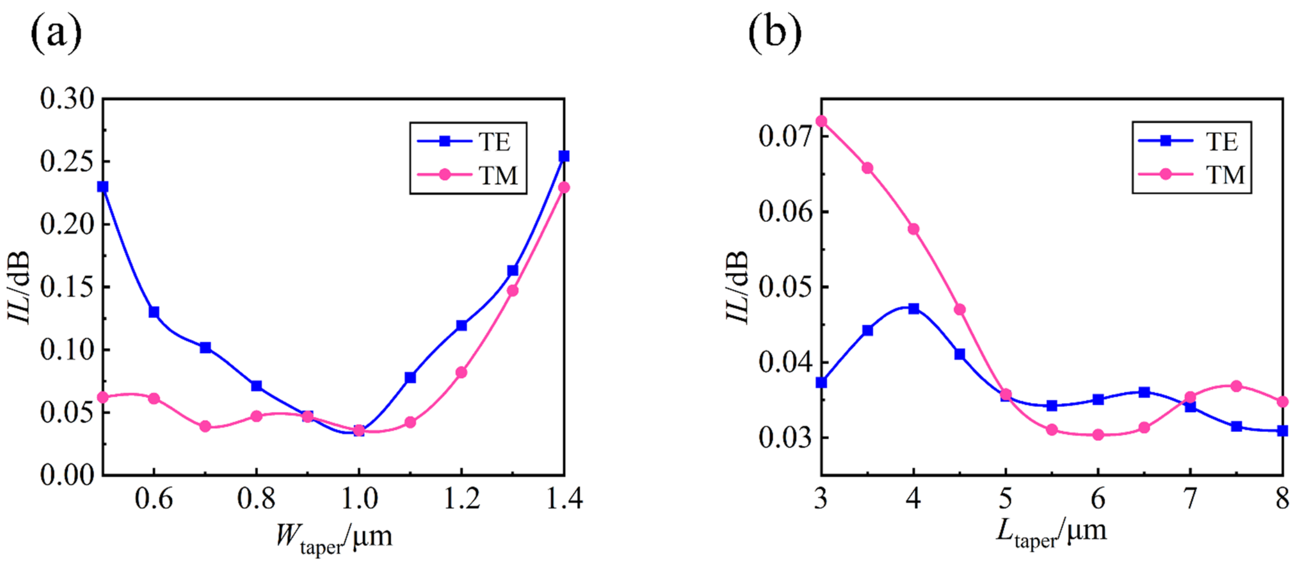

Fig. 6. (**a**) The relationship between insertion loss *IL* and taper width W_{taper}, which first decreased and then increased with the increase of W_{taper} (**b**) the relationship between insertion loss *IL* and taper length L_{taper}, which generally showed a decreasing trend with the increase of L_{taper}

In summary, the optimized device parameters are as follows: W_{MMI} =4 μm, L_{MMI} =15.6 μm, $n(SiN_x)=2.51$, $w_1=0.5$ μm, W_{taper} =1 μm, and L_{taper} =5.5 μm. The device successfully performs a 1 × 2 optical signal splitting function at a wavelength of 1550 nm. Additionally, the two polarization modes, TE and TM, exhibit minimal to no effect on the output, thereby ensuring polarization independence and optimal device performance. Figure 7 displays the optical field distribution of the signal propagating through the polarization-independent MMI 1 × 2 optical power splitter.

Preparative tolerance and performance analysis

During the manufacturing process, the actual dimensions of the device may deviate from the calculated values due to variations in the manufacturing technology and process, which in turn affects the device performance. To ensure the device meets the required performance standards in practical applications, tolerance analysis of the device parameters is essential. The fabrication tolerance is directly linked to the device performance, reliability,

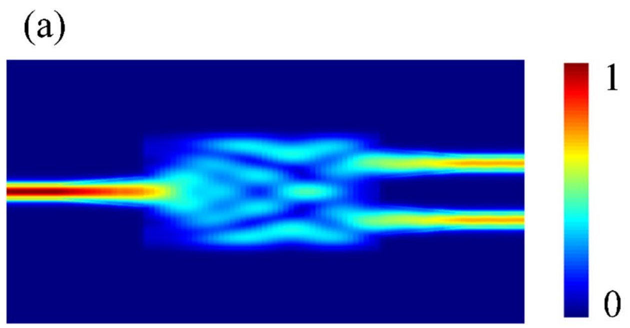

Fig. 7. Optical field distribution of polarization-independent MMI type 1 × 2 optical power splitter (**a**)TE (**b**) TM.

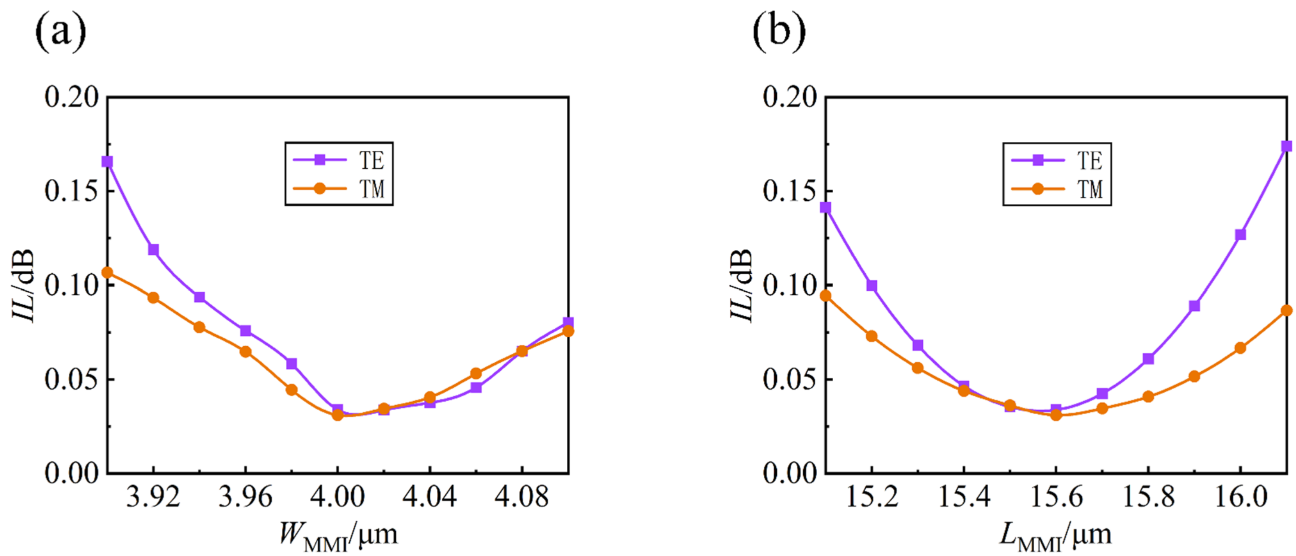

Fig. 8. The larger the fabrication error of the device, the greater the insertion loss. (**a**) The change relationship between *IL* and W_{MMI} when W_{MMI} changes from 3.9 to 4.1. (**b**) The change relationship between *IL* and L_{MMI} when L_{MMI} changes from 15.1 to 16.1.

and manufacturing cost. A reasonable preparation tolerance ensures that the device can maintain the design-required performance standards even with manufacturing errors, thereby reducing performance degradation due to dimensional variations.

The device tolerance has been analyzed, focusing on the impact of variations in W_{MMI} and L_{MMI} on the insertion loss *IL* while keeping other parameters constant. As shown in Fig. 8a, when W_{MMI} varies between 3.9 and 4.1 μm, the *IL* of the TE mode remains below 0.17 dB, and the *IL* of the TM mode stays below 0.11 dB, indicating good tolerance of the device.

As shown in Fig. 8b, the TE mode is more sensitive to preparation errors than the TM mode. The figure demonstrates that when L_{MMI} changes within the range of 15.1 to 16.1 μm, the *IL* of the TE mode remains below 0.175 dB, while the *IL* of the TM mode stays below 0.1 dB, further confirming the device good preparation tolerance.

In practical applications, since the optical source is not monochromatic, it is essential to consider the impact of bandwidth on the device's performance. According to Literature[26], the refractive index of SiN_x material in the infrared band exhibits strong stability with respect to wavelength variations, indicating that within this range, the refractive index of SiN_x remains virtually constant across different wavelengths. This property ensures that the device's performance remains consistent across various polarization states, without compromising its polarization-independent characteristics. Figure 9 shows the relationship between insertion loss and wavelength variation within the wavelength range of 1510 nm to 1610 nm. In the bandwidth range (1510 nm to 1610 nm), the loss of both polarization modes of the device is below 0.175 dB.

The aforementioned MMI coupler achieves both polarization-independent and 1 × 2 optical power splitter function. After simulation and optimization, the specific parameters of the power splitter are as follows: The Si layer thickness is 0.25 μm, the thickness of SiN_x of the core layer is 0.1 μm, the coupling region length is 15.6 μm, the coupling region width is 4 μm, the cone width is 1 μm, and the cone length is 5.5 μm. The TE polarization

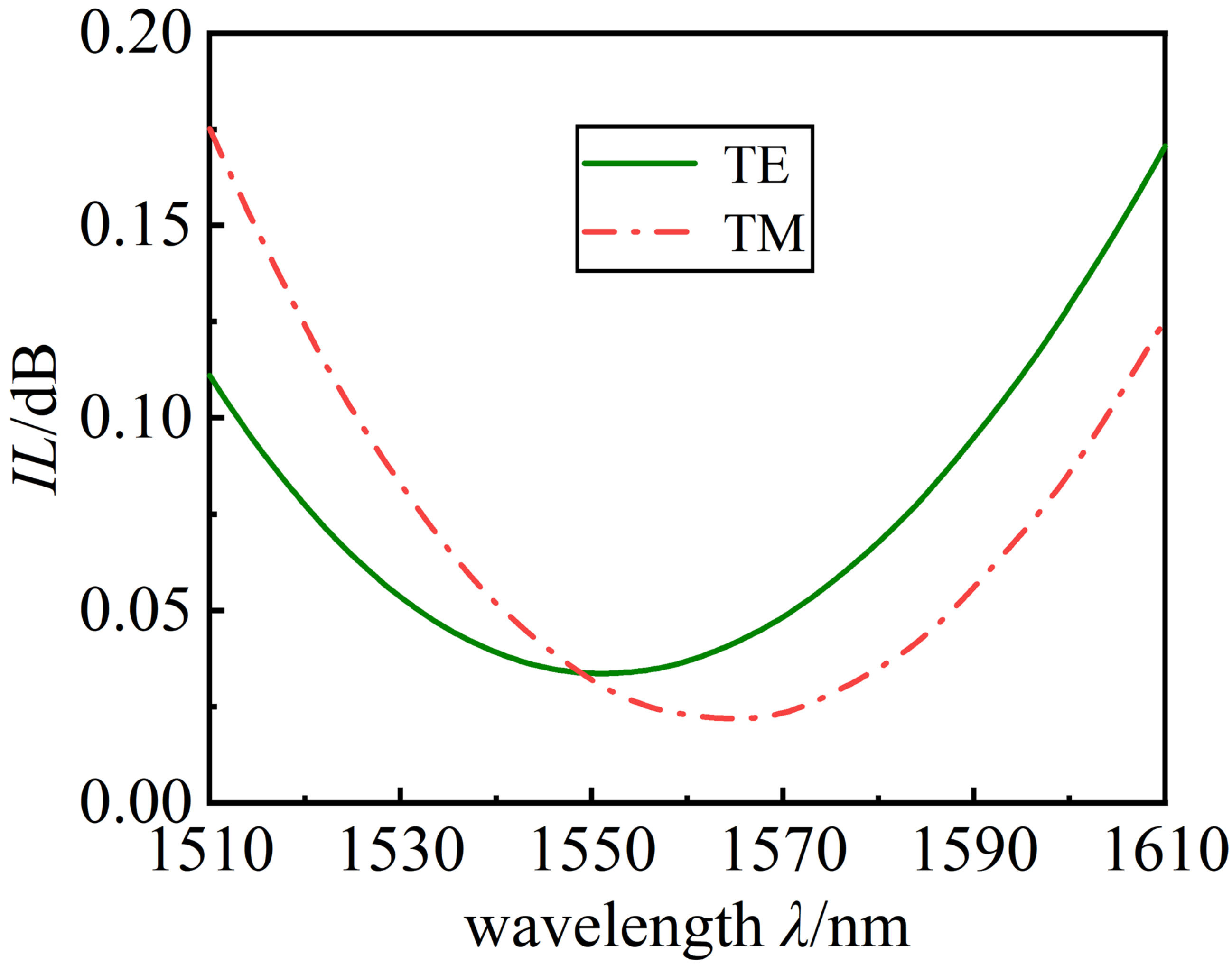

Fig. 9. Relation of insertion loss *IL* with wavelength λ, the solid green line represents TE mode, the dotted red line represents TM mode.

Ref.	Footprint (μm2)	Type	ILTE (dB)	ILTM (dB)	Bandwidth (nm)
[7]	0.93 × 5	Y-junction	0.11	0.08	1530–1600(*IL*<0.2 dB)
[8]	1.3 × 7	DC	0.04	0.02	1520–1620(*IL*<0.31 dB)
[14]	5 × 24.54	MMI	0.66	0.86	1500–1600(- dB)
[17]	3.8 × 15.2	MMI	0.12	0.23	1530–1590(*IL*<0.5 dB)
[18]	1.2 × 2.62	DC	0.18	0.35	1540–1590(*IL*<1 dB)
[19]	2 × 1.92	MMI	0.39	0.39	1510–1615(*IL*<0.77 dB)
This work	4 × 15.6	MMI	0.034	0.031	1510–1610(*IL*<0.175 dB)

Table 1. Comparison of performance parameters of polarization-independent optical power splitter.

mode loss of the device at a 1550 nm wavelength is less than 0.034 dB, and the TM polarization mode loss is less than 0.031 dB. The device also exhibits a bandwidth of up to 100 nm while maintaining *IL* below 0.175 dB.

Table 1 presents a comparison of various parameters for different types of polarization-independent optical power splitter. As shown, although the devices in references[7,17,18] have a more compact structure, their bandwidth is smaller. In contrast, the present work offers a wider bandwidth and lower loss. Compared to references[8,19], the loss in this paper is lower while maintaining a similar bandwidth. Furthermore, when compared to reference[14], the device in this study is both smaller in size and exhibits lower loss.

Conclusion

This study presents the design of a polarization-independent 1 × 2 optical power splitter based on MMI. The device utilizes a three-layer slot waveguide structure composed of Si/ SiN_x /Si to achieve polarization-independent function. To accomplish this, the refractive index of SiN_x is adjusted using PECVD, ensuring that the beat length of the two polarization modes is equal at the same wavelength. The variations of L_{MMI} with respect to $n(SiN_x)$ and W_{MMI}, as well as the impact of W_{MMI} on IL, are subsequently analyzed. Under polarization-independent conditions, a W_{MMI} with low loss and small size is selected. Next, a taper structure is simulated, and by optimizing the taper length L_{taper} and width W_{taper}, the insertion loss is effectively reduced. The results show that the insertion loss for the TE and TM polarization modes is below 0.034 dB and 0.031 dB, respectively, at a wavelength of 1550 nm. Additionally, the insertion loss for both polarization modes remains below 0.175 dB across the 100 nm bandwidth range from 1510 nm to 1610 nm. The device demonstrates excellent performance, a straightforward design, and holds great potential in photonic integrated circuit (PIC) technology.

Data availability

Data supporting this study are available from the corresponding author on reasonable request.

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References

1. Dai, D., Bauters, J. & Bowers, J. E. Passive technologies for future large-scale photonic integrated circuits on silicon: Polarization handling, light non-reciprocity and loss reduction. *Light Sci. Appl.* **1**, e1–e1 (2012).
2. Yin, Y. et al. A Silicon-Based On-Chip 64-Channel hybrid Wavelength- and Mode-Division (de)Multiplexer. *Photonics* **10** (2023).
3. Chen, X., Lin, J. & Wang, K. A review of Silicon-Based integrated optical switches. *Laser Photonics Rev.* **17** (2023).
4. Kim, M., Kim, S. & Kim, S. Graphene optical modulators using bound States in the continuum. *Sci. Rep.* **12**, 1445 (2022).
5. Li, L. et al. Switchable 3D optofluidic Y-branch waveguides tuned by dean flows. *Sci. Rep.* **6** (2016).
6. Song, C. et al. Ultracompact and broadband Si3N4 Y-branch splitter using an inverse design method. *Opt. Express*. **32**, 46080–46089 (2024).
7. Wang, Y., Gao, S., Wang, K. & Skafidas, E. Ultra-broadband and low-loss 3 dB optical power splitter based on adiabatic tapered silicon waveguides. *Opt. Lett.* **41**, 2053–2056 (2016).
8. Wang, J. et al. Polarization-Independent optical power splitter with a designable splitting ratio. *Laser Optoelectron. Progress* **60** (2023).
9. Dash, R. K. & Selvaraja, S. K. Passive wavelength selective polarization rotator in a hybrid waveguide platform. *Opt. Express*. **32**, 13035–13047 (2024).
10. Soldano, L. B. & Pennings, E. C. M. Optical multi-mode interference devices based on self-imaging: Principles and applications. *J. Lightwave Technol.* **13**, 615–627 (1995).
11. Qiyuan, Y. et al. Silicon MMI-based power splitter for multi-band operation at the 1.55 and 2 μm wave bands. *Opt. Lett.* **48**, 1335–1338 (2023).
12. Rene, J. B. D., Douwe, H., Arne, G. & Chris, R. L. Silicon nitride in silicon photonics. *Proc. IEEE* **105**, 2209–2231 (2018).
13. Xiang, C., Jin, W. & Bowers, J. E. Silicon nitride passive and active photonic integrated circuits: Trends and prospects. *Photonics Res.* **10** (2022).
14. Sun, X., Alam, M. Z., Aitchison, J. S. & Mojahedi, M. Compact and broadband polarization beam splitter based on a silicon nitride augmented low-index guiding structure. *Opt. Lett.* **41**, 163–166 (2016).
15. Sharma, T. K., Ranganath, P., Nambiar, S. & Selvaraja, S. K. Broadband transverse magnetic pass polarizer with low insertion loss based on silicon nitride waveguide. *Opt. Eng.* **57**, 037104–037104 (2018).
16. Rahman et al. Compact polarization-independent MMI-based 1 × 2 power splitter using metal-cap silicon-on-insulator waveguide. *IEEE Photonics J.* **8**, 1–1 (2016).
17. Dai, D. & He, S. Optimization of ultracompact polarization-insensitive multimode interference couplers based on Si nanowire waveguides. *IEEE Photonics Technol. Lett.* **18**, 2017–2019 (2006).
18. Aoxue, Z. et al. Ultra-compact polarization-independent 3 dB power splitter in silicon. *Opt. Lett.* **46**, 5000–5003 (2021).
19. Xu, Y. & Xiao, J. An ultracompact polarization-insensitive silicon-based strip-to-slot power splitter. *IEEE Photonics Technol. Lett.* **28**, 536–539 (2016).
20. Vogelbacher, F. et al. Slot-Waveguide silicon nitride organic hybrid distributed feedback laser. *Sci. Rep.* **9**, 18438 (2019).
21. Li-Min, C. et al. Polarization-independent directional coupler and polarization beam splitter based on asymmetric cross-slot waveguides. *Appl. Opt.* **57**, 678–683 (2018).
22. Katash, N., Khateeb, S. & Malka, D. Combining four Gaussian lasers using silicon nitride MMI slot waveguide structure. *Micromachines (Basel)* **13** (2022).
23. Wang, J., Huangfu, L., Chen, H. & Zhong, K. Design of ultra-compact and polarization-insensitive multimode interference demultiplexer. *Opt. Communi* **500** (2021).
24. Zhaoming, Z. & Thomas, B. Full-vectorial finite-difference analysis of micro structured optical fibers. *Opt. Express*. **10**, 853–864 (2002).
25. Hansbo, P. & Rylander, T. A linear nonconforming finite element method for maxwell's equations in two dimensions. Part I: Frequency domain. *J. Comput. Phys.* **229**, 6534–6547 (2010).
26. Guler, I. Optical and structural characterization of silicon nitride thin films deposited by PECVD. *Mater. Sci. Eng. B* **246**, 21–26 (2019).

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Author contributions

Fangxv Liu and Huanlin Lv conceived the idea. Fangxv Liu carried out the theoretical analysis, simulation and wrote the main manuscript text. All authors discussed the results and contributed to the writing of the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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